

Acoustic Detection of Impulsive Sounds

[0:00–0:45]

Good morning. // I'm Martin Maxa from the Department of Measurement, here at this faculty. // This is joint work with Jakub Svatoš from the same department. //

Today I'll talk about detecting, classifying and localizing impulsive sounds – in our case, gunshots. //

The main part is gunshot classification, where we compare different cepstral features and two classifiers. // And I'll briefly show how the system localizes the source. //

(// = pause. [CLICK] = advance the slide.)

Acoustic Detection of Impulsive Sounds

└ Motivation

└ Motivation and research questions

Motivation and research questions

- Acoustic detection of gunfire complements camera-based surveillance: the acoustic channel is omnidirectional and does not require line of sight.
- Three required functions, in outdoor environments that are noisy and reverberant:


detection
of the impulsive event


classification
ideally down to calibre


localization
of the source, in 2D

This study

1. Which cepstral representation separates gunshots from acoustically similar false alarms most reliably?
2. Neural network or support vector machine on those features?
3. How does the analysis frame length interact with environmental reflections?

[0:45–1:50] (1/2)

This work is part of a larger project – an acoustic sensor system for protecting soft targets. // Acoustic detection is a good complement to cameras. // Sound is omnidirectional, and you don't need a line of sight. //

[CUT IF LATE] The full system has to detect an impulsive event, classify it – ideally down to the calibre – and localize the source. // And it has to work outdoors, with noise and a lot of reflections.

[/CUT] //

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[0:45–1:50] (2/2)

[CLICK]

In this study we ask three questions. //

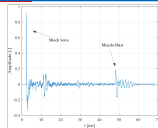
First, which cepstral features best separate gunshots from similar impulsive sounds? // Second, which classifier works better – a neural network or a support vector machine? // And third, how does the frame length affect the results? //

Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



One or two components, depending on projectile speed:

- **Shock wave** ("N-wave") - supersonic only ($v > c$); generated along the trajectory, so it arrives first.
- **Muzzle blast** - always present; propellant gases at the barrel, 1-3 ms; here ~45 ms behind.
- Between and after: **reflections**.

[1:50-3:00] (1/2)

Let's start with the signal of a gunshot. //

A gunshot has one or two components, depending on the projectile speed. // A subsonic shot has only the muzzle blast. // A supersonic one also has a shock wave. // This recording is a supersonic shot, so we can see both. //

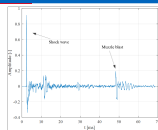
The first one, at about two and a half milliseconds, is the shock wave. // [CUT IF LATE] It's created along the bullet's path and has the typical N-shape. [/CUT] //

Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



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[1:50-3:00] (2/2)

[CLICK]

The second one, about forty-five milliseconds later, is the muzzle blast. // It comes from the gases leaving the barrel, and every gunshot has it. //

Everything between and after them is mostly reflections from the surroundings. // This matters for the frame length, and I'll come back to it in the results. //

Acoustic Detection of Impulsive Sounds

└ Signal

└ Single-domain analysis is insufficient

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

Effective feature extraction must integrate both time and frequency domain processing.

[3:00–3:30]

This also shows why one domain alone is not enough. //

The spectrum often tells you more about the environment than about the gunshot. // And the waveform changes as the sound travels and hits obstacles. //

So we need features that combine both domains, and cepstral features do exactly that. //

Acoustic Detection of Impulsive Sounds

└ System

└ Proposed system architecture

Proposed system architecture



[3:30–4:20]

Now to the system itself. // It's a set of independent sensor units connected to a remote server. //

Each unit continuously monitors its surroundings. // When it detects an impulsive event, it sends the recorded signal to the server. // The server then does two things – it classifies the event and localizes the source. //

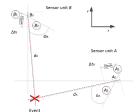
More units mean wider coverage, and they let us estimate the position of the source. //

We tested the system with several calibres and types of ammunition. //

Acoustic Detection of Impulsive Sounds

└ System

└ Localization – bearing from TDoA



$$\Theta = \arcsin\left(\frac{c \Delta t}{d}\right)$$

- Two MEMS microphones per unit ($d = 0.186\text{ m}$, $f_s = 44.1\text{ kHz}$) give one bearing Θ .
- Two units, two bearings – they intersect at the event.
- Measured: cross-correlation with parabolic interpolation, **MAE 0.77°** over 70 events (-45° to $+60^\circ$).

[4:20–5:20] (1/2)

Let me briefly explain the localization first. //

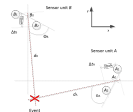
Each sensor unit has two MEMS microphones, about twenty centimetres apart, sampled at 44.1 kilohertz. //

We measure the time difference of arrival between the two microphones. // From this delay we get the direction to the source, using the equation here. //

Acoustic Detection of Impulsive Sounds

System

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[4:20–5:20] (2/2)**[CLICK]**

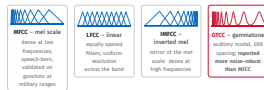
So one unit gives us one direction. // With two units we get two directions, and where they cross is the position of the source. //

We tested this on seventy events at nine reference angles. // With cross-correlation and parabolic interpolation, the mean absolute error was zero point seven seven degrees. //

Acoustic Detection of Impulsive Sounds

Features

Feature extraction – four cepstral representations



Experimental grid: 4 extractors $\times$ 3 frame lengths (75 / 100 / 150 ms) $\times$ 2 classifiers (NN, SVM).

[5:20–5:55]

Now to the main part – classification. //

We compared four cepstral feature extractors. // They differ in the filter bank they use. //

MFCC uses the mel scale, with dense filters at low frequencies. // It comes from speech processing, but it works for gunshots too. //

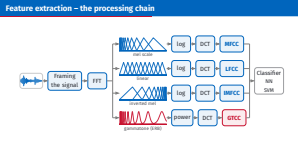
LFCC has linearly spaced filters, and IMFCC uses an inverted mel scale, dense at high frequencies. //

GTCC uses a gammatone filter bank, which is based on a model of human hearing. // It's also reported to be more robust to noise than MFCC. //

Acoustic Detection of Impulsive Sounds

Features

Feature extraction – the processing chain



[5:55–6:25]

And this is how the features are computed. //

All four use the same chain – framing, FFT, the filter bank, then a logarithm and a DCT. // The filter bank is the only place where they differ, and here you can see all four of them on the same frequency axis. //

The coefficients then go to a classifier. // We tested two – a neural network and a support vector machine – with three frame lengths, fifteen, thirty and fifty milliseconds. //

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Results

Classification results – 9 mm class

Classification results – 9 mm class

Frame	Extractor	ACC (%)		BLC (%)		MCC (-)	
		SVM	NN	SVM	NN	SVM	NN
15ms	MFCC	83.90	85.59	80.65	82.76	0.62	0.62
	LFCC	83.90	83.90	80.65	84.52	0.62	0.57
	MFCC	85.59	90.68	83.87	87.92	0.66	0.77
	GTCC	93.22	97.66	87.98	93.95	0.82	0.90
30ms	MFCC	86.75	87.29	82.76	87.92	0.60	0.70
	LFCC	87.29	83.90	79.87	79.87	0.66	0.59
	MFCC	85.59	92.37	82.76	89.65	0.62	0.80
	GTCC	97.66	98.21	93.22	93.95	0.93	0.96
50ms	MFCC	86.75	92.22	76.19	90.32	0.61	0.83
	LFCC	88.08	86.44	77.42	77.42	0.71	0.66
	MFCC	88.83	90.68	76.19	89.65	0.73	0.76
	GTCC	97.66	98.95	93.22	96.77	0.93	0.98

Classification results for the 9mm class. Test partition: 718 events, 31 of them 9mm.

[6:25–6:40]

This is the full table of results for the nine-millimetre class. // The GTCC rows are highlighted. //

I won't go through the numbers – let's look at the same data as a chart. //

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Results

Classification results – MCC at a glance

**[6:40–7:35] (1/2)**

Here are the same results, just the Matthews correlation coefficient. // MCC takes the whole confusion matrix into account, so it's less affected by class imbalance than accuracy. //

The red line is GTCC. // It's on top in both panels, at every frame length. //

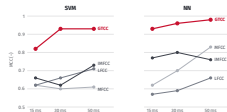
The neural network does better than the SVM, and the gap grows with longer frames. //

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Results

Classification results – MCC at a glance

Classification results – MCC at a glance



MCC for the train class by frame length – the same data as the table (708 test events, 21 of them 9 ms).

[6:40–7:35] (2/2)

The best configuration is GTCC with the neural network and a fifty-millisecond frame. // That gives ninety-nine point one five percent accuracy, ninety-six point seven seven percent recall, and an MCC of zero point nine eight. //

In other words, one hundred and seventeen correct decisions out of one hundred and eighteen. //

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Results

Discussion

- MFCC, LFCC and IMFCC yield similar results.
- GTCC consistently performs better across every frame length and both classifiers.
- NN outperforms SVM - better at modelling complex patterns, especially with longer frames.
- Longer frames improve accuracy, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms - 99.15 % ACC, 96.77 % RLC, MCC 0.98.

[7:35-8:45] (1/2)

Three things to take from this. //

First, MFCC, LFCC and IMFCC give similar results, so the filter spacing makes little difference. //

Second, GTCC is clearly better. // [CUT IF LATE] One likely reason is that the gammatone bank comes from a hearing model and is known to be more robust to noise. // But we didn't test that directly.

[/CUT] //

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[7:35-8:45] (2/2)

Third, the neural network is slightly better than the SVM, especially with longer frames. // That's a general trend, not true for every single result. //

[CLICK]

And finally, longer frames generally improve accuracy. // But at fifty milliseconds, the frame already contains more and more reflections, not just the gunshot itself. // So even longer frames wouldn't necessarily help. //

Acoustic Detection of Impulsive Sounds

└ Conclusion

└ Conclusion

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 20.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you – questions?

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[8:45–9:30]

To sum up. //

We presented a distributed acoustic sensor system that detects, classifies and localizes impulsive sounds. //

For gunshot classification, cepstral features work well with both classifiers. // GTCC gives the best results of the four. // And the neural network is slightly better than the SVM, especially with longer frames. //

All of this was tested on real firearms, with several calibres and types of ammunition. //

Thank you. //